



CLIFF CASEY

CTO & SOFTWARE ARCHITECT

CONTACT



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Baltimore, MD

PROFILE

Extensive experience hiring, training, managing and coordinating engineering teams (internal and offshore), vendors, contractors, infrastructure teams and QA resources. Adept at effectively communicating technical concepts and risks to stakeholders. Have overseen large staff transitions, corporate pivots and acquisitions. Proven track record of understanding business needs, creating innovative and appropriately delivered solutions, and steering organizational direction and strategy from a technical perspective.

Decades of experience building scalable enterprise applications and applying best practice design principles and patterns to software and infrastructure. Skilled at relational, document and graph persistence architecture, implementation and optimization. Experience implementing SCRUM, Agile, Waterfall and other common SDLCs and related project management, source control and code review practices. Proven track record of successfully migrating legacy applications to the cloud as well as building cloud-native solutions in PaaS providers such as AWS. Directed responsible integration of LLM use into developer workflows, increasing productivity without sacrificing product quality or technical debt.

Sector-specific software engineering and architecture experience in Federal Government / DoD, education data, e-commerce, marketing, publishing, NPOs, healthcare, IT infrastructure management, workflow / process management and logistics.

SKILLS & TECHNICAL SUMMARY

Core Languages

Java, Clojure, Js/Ts, Python, C/C++, PHP, C#

Persistence

PostgreSQL, SQL Server, MySQL / Maria, SQLite, Graph & Document DBs, K-V Stores, Data Warehousing

Frontend

React (+Native), HTML5, jQuery, CSS/SASS

Platforms

Tomcat, JBoss, Jetty, Apache, Nginx, Linux

Machine Learning

LLM Code Analysis, AI Policy, ML Classification and Regression Models

Containerization & IaC

Docker, Kubernetes, CloudFormation, ArgoCD

Cloud Infrastructure

AWS (Robust knowledge and experience, CloudFormation best practices), Azure, PlatformOne

Collaboration Tools

Git, Jira, Asana, SVN, Confluence

CI/CD

GitHub Actions, Jenkins, Bamboo, BitBucket Pipelines

Game Engines

Unreal Engine 5 (C++ & BP), Unity, LibGDX

Security

RBAC, LDAP/AD, SSO/SAML, OAuth 2.0, TLS, Secure SDLC practices, PCI, SOC 2, NIST CSF & 800-171, OWASP

Protocols & Formats

JSON, REST, XML / SOAP, YAML, EDN

PROFESSIONAL EXPERIENCE

Chief Technology Officer

Yet Analytics Inc, Baltimore MD | Sep 2019-Present

Leading the technical side of an education technology software company to deliver best-in-class learning data solutions to commercial and DoD clients. Responsibilities include hiring and managing technical resources, architecture and engineering of software and cloud infrastructure, managing client technical relationships and projects, and directing and advising the company as a whole. Redeveloped most company software products and transitioned the company to an Open Source

software model in 2021. Fulfilled numerous federal government research and development contracts in the education and simulation space as Principal Investigator (PI). Serve on related IEEE committees and working groups, shaping global learning data standards (LTSC Leadership, P2881, P2997, 9274) and acting as liaison between the IEEE Learning Technology Standards Committee (LTSC) and Simulation Interoperability Standards Organization (SISO).

Primary technologies include Java, Clojure, Javascript, React, PostgreSQL, Datomic, and xAPI. Deploy systems on AWS, Azure and custom / DoD deployment environments. Maintain secured hardened DoD versions of software in high security environments. Deployed custom ML algorithms for event analysis and content recommendations.

Notable clients/projects have included the US DoD Advanced Distributed Learning Initiative, US Navy NETC, US Air Force, US Space Force, US Army STTC, Deloitte, Boeing, IBM, HTX, Moodle, NYU, Verizon, American Red Cross, the New Zealand Defence Force, the Netherlands Armed Forces.

Chief Technology Officer

SafetyPIN Technologies Inc, Baltimore MD | Jan 2018-Sep 2019

Architected and built the core systems for a B2C startup in the background check and ID verification space. Responsibilities included development and maintenance of the core product, technical platform and vendor selection and integration for marketing and peripheral technologies, hiring and managing development and QA resources, contractor and vendor management, enterprise partner integration, code reviews, security analysis, and directing and advising the company from a technical perspective.

Core technologies included Java, SQL, AngularJS, and a heavy reliance on AWS IaaS services such as EC2, RDB, ElastiCache, SQS and SES. Designed and developed a machine learning classification models which automated the evaluation of parts of background check applications, improving processing speed and consistency. Developed mobile companion applications to the primary system for Android and iOS in Ionic/Cordova.

Principal Engineer

South Ferry IT Consulting LLC, Baltimore MD | Nov 2010– Sep 2019

Founded a consulting company in 2010 to execute projects for various clients and media agencies in roles such as Lead Developer, Technical Director, Advisor, and Interim CTO. Built applications for client needs across many platforms and industries. Clients have included Norfolk Southern, Washington Theological Consortium, Colorado Donors Alliance, Staffing Advisors, Adboku, Interfaith Conference of Washington, IRN, Catholic Volunteers Network, Arbit Inc, StudioRHM, Chinese Language Institute, Perisphere Media, and the University of California Riverside.

Chief Solutions Architect

Publishing Services, Agora Inc, Baltimore MD | Dec 2013- Dec 2017

The Chief Solutions Architect role included performing reviews of all technology choices over the entire range of products and becoming an advocate for improving or replacing legacy systems and enhancing the company's enterprise stack by taking advantage of the latest advances in the industry. It also included the design, proposal and development of new systems, hiring and building new teams and performing code reviews and training.

The Chief Solutions Architect advised the organization on upcoming projects, new technologies, SDLC process, and new opportunities. The role was also responsible for technical interviews and training, and assisted in recruiting and education events in the Baltimore technical community.

Lead Engineer (E-Commerce)

Publishing Services, Agora Inc, Baltimore MD | Jan 2012– Dec 2013

Led a team of Java developers to build and expand an enterprise E-Commerce gateway and marketing suite, and provide direction on many projects for a multi-national publishing consortium. Transitioned the development team from consultants to in-house developers under this role's leadership and training. The role included system architecture, traditional development responsibilities, and the hiring, training and guidance of many junior and mid-level developers. The role also consisted of code reviews, coordinating sensitive production deployments and migrations, and communicating effectively with product owners, project managers, QA, support and the clients themselves.

Lead Application Developer

Government Services IPT, Largo MD | Jan 2010–Jan 2012

Worked as a software engineer for a Government contractor building applications for Federal / DoD clients across a wide range of requirements and platforms. Built Project Management / ERP solutions, EVM tools and FCA (Facility Condition Assessment) software. Specific technologies included Java, PHP, SQL. Worked as part of a small SCRUM team and integrated agile development methodologies in day-to-day work. Notable clients included NASA and the US Army Corps of Engineers (USACE).

Developer / Wireless Systems Planner

NCS, McGill University, Montreal QC, Canada | May 2008–July 2009

Started as a deployment and planning specialist designing an updated wireless service infrastructure across numerous campuses. Transitioned into applications development and worked in an MRTG / SNMP / PHP / Perl / Bash environment writing applications to provide network data for real-time monitoring of tens of thousands of devices across large campus networks in the Montreal area. Job was performed while in pursuit of a B. Engineering.

Application Developer

Antharia LLC, Seabrook MD | June 2001–Dec 2007

Developer for a successful media studio, contributing to a diverse line of browser-based software suites utilizing PHP, MySQL and Js. Products included a comprehensive custom CMS, CRM, and numerous other tools and the clients consisted primarily of NPOs in the DC metro area. The position fluctuated between part-time and full-time based on education commitments. Some notable clients included the ACLU and Discovery Channel.

EDUCATION

Bachelor of Engineering (Computer)

McGill University, Montreal, QC | 2009

(Note: U.S. Citizen)

PUBLICATIONS

*Robson, R., Goldberg, B., Blake-Plock, S., Casey, C., Hoyt, W., Hernandez, M., Ray, F. (2022). **Mining Artificially Generated Data to Estimate Competency**. Proceedings of the 15th International Conference on Educational Data Mining. Durham University. Durham, UK.*

*Owens, K., Goldberg, B., Robson, R., Hoffman, M., Ray, F., Colburn, A., Hernandez, M., Blake-Plock, S., Casey, C. (2022). **Synthetic Training Environment Experiential Learning for Readiness (STEEL-R)**. Proceedings of the Tenth Annual GIFT Users Symposium. US Army Combat Capabilities Development Command - Soldier Center. Orlando, FL.*

Blake-Plock, S., Hoyt, W., & Casey, C. (2021). **Instrumenting GIFT with xAPI: a use case for IEEE P9274.3.x standards activity and implications for the broader field of ITS and AIS**. *Proceedings of the Ninth Annual GIFT Users Symposium*. US Army Combat Capabilities Development Command - Soldier Center. Orlando, FL.

Blake-Plock, S., Hoyt, W., Casey, C., & Zapata-Rivera, D. (2020). **Data Analytics and Visualization for xAPI Learning Data: Considerations for a GIFT Strategy**. In Sinatra, A.M., Graesser, A.C., Hu, X., Goldberg, B., and Hampton, A.J. (Eds.). *Design Recommendations for Intelligent Tutoring Systems: Volume 8 - Data Visualization*. US Army Combat Capabilities Development Command - Soldier Center. Orlando, FL.

Blake-Plock, S. and Casey, C. (2020). **DATASIM and xAPI Profile Validation**. *iFEST 2020*. National Training and Simulation Association. Alexandria, VA.

NOTABLE OPEN SOURCE CONTRIBUTIONS

[SQL LRS](#) - Clojure

Apache 2.0 Open Source Learning Record Store (LRS) with support for SQLite, PostgreSQL and MySQL/MariaDB. Created this product in 2020 to pivot Yet Analytics to an Open Source software model, developed it with Yet engineering team, and it became one of the most widely used LRS implementations in the world.

[Enterprise Learner Record Repository \(ELRR\)](#) - Java/Spring

Apache 2.0 Open Source IEEE P2997 Learner Record Repository. Developed in collaboration with US DoD and Deloitte.

[DATASIM](#) - Clojure

Apache 2.0 Open Source Synthetic xAPI Data Generator. Developed in collaboration with US DoD.

[Moodle xAPI Logstore](#) - PHP

Contributed to a 2024 overhaul of a widely used xAPI implementation for one of the most popular Learning Management Systems, enabling comprehensive xAPI instrumentation.

[Java xAPI Tools](#) - Java

Small library for handling xAPI data in Java, complete with conformant client. Published on Maven Central.

[LaneGraph](#) - C++ (Unreal)

System for generating virtual n-lane traffic corridors from splines which heavily influences navigation algorithms without restricting movement. In practice it makes simulated traffic seem more organic and able to deal with emergent obstacles. Includes A* nav implementation.

[SplintFormation](#) - C++ (Unreal)

Toolkit primarily for extracting and manipulating information from Landscape Splines which are locked down and missing a robust API in Unreal. Was used for procedurally generated buildings along a landscape-created spline path/road and includes some helpers for exactly packing splines with stacked multi-length objects (buildings, in this case).